1. RtHDDump Content Difference Report

1.1. Abstract

This report investigates RtHDDump file **content** differences and their relationship to device states (preferred device, headset plug state, and Dolby/enhancement modes). The analysis focuses on WID lines and registry-style key/value entries inside each dump rather than relying on filename tokens. We find that WID lines are identical across all dumps, while the state changes consistently align with differences in specific `REG_*` keys.

1.2. Data and methods

- Dataset: 27 RtHDDump captures.
- WID analysis: extract all lines starting with `Wid=` and compare across files.
- Content difference analysis: extract key/value lines (`<key> = <value>`).
- Paired comparison method: for each device state, match files with all other states held constant and compare their key/value lines. A key is a **signature** when it changes in ≥ half of the paired comparisons for that state.
- Values are trimmed for readability (`prefix…suffix`) but still show the changing portions.

1.3. Direct answer (WID vs device states)

No WID lines change across the dataset, so there is no WID that directly controls Dolby, preferred device, or system audio enhancement in these dumps. The state changes are instead reflected in `REG_*` key/value differences:

State	WID control?	Content keys that change (examples)
Dolby (speaker/headset)	None observed	`(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`, `(REG_BINARY) {8a845654-d6c3-4cd7-b4eb-243d4bd99032},2`, `(REG_BINARY) {6737016f-5360-48ee-af05-e616c8ff27a7},2`, `(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4`
Preferred (primary) device	None observed	`(REG_SZ) {24dbb0fc-9311-4b3d-9cf0-18ff155639d4},0`, `(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`, `(REG_BINARY) {bb8bdb4a-edac-4660-9056-8e67e68e4e77},4`
System audio enhancement	None observed	`(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`, `(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1`, `(REG_DWORD) {1da5d803-d492-4edd-8c23-e0c0ffee7f0e},5`

1.4. Linux codec dump analysis

Linux dumps (`lin_` prefix) are HD-audio codec snapshots (Realtek ALC287). Their tokens describe ALSA mixer states:

- `no_automute`: automute disabled via alsamixer.
- `dualstream`: manual unmute of both headphone and speaker paths while the headset is plugged.
- `dual` (in the filename): headset is plugged.

The Linux dumps show a single stream shared by the speaker/headphone outputs (node 0x02 and 0x03). The speaker pins (node 0x14/0x17) switch between `0x80` (muted) and `0x00` (unmuted) when automute/dualstream changes.

File	dual (hp plugged)	no_automute	dualstream	node0x02 stream	node0x03 stream	node0x14 amp-out	node0x17 amp-out
lin_codec-dump-dual_h_spk	yes	no	no	0	0	0x80 0x80	0x80 0x80
lin_codec-dump-dual_h_spk_no_automute	yes	yes	no	1	1	0x80 0x80	0x80 0x80
lin_codec-dump-dual_h_spk_no_automute_dualstream	yes	yes	yes	1	1	0x00 0x00	0x00 0x00
lin_codec-dump-spk	no	no	no	0	0	0x00 0x00	0x00 0x00
lin_codec-dump-spk_no_automute	no	yes	no	1	1	0x00 0x00	0x00 0x00
lin_codec-dump-spk_no_automute_dualstream	no	yes	yes	1	1	0x00 0x00	0x00 0x00

Interpretation: `dualstream` explicitly unmutes the speaker pins while the headset is plugged, so the same audio stream is audible from both ports. `no_automute` turns off ALSA automute but does not by itself unmute the speaker pins when the headset is plugged; the manual unmute (dualstream) is what clears the `0x80` mute values.

1.4.1. Linux coefficient (verb) deltas

The refreshed Linux dumps now include the vendor coefficient block under Node 0x20. Only three coefficients differ across the Linux states:

File	dual	no_automute	dualstream	Coeff 0x46	Coeff 0x77	Coeff 0x78
lin_codec-dump-dual_h_spk	yes	no	no	0c04	0063	0079
lin_codec-dump-dual_h_spk_no_automute	yes	yes	no	0c04	0023	0090
lin_codec-dump-dual_h_spk_no_automute_dualstream	yes	yes	yes	0c04	0046	005d
lin_codec-dump-spk	no	no	no	0404	0050	00a6
lin_codec-dump-spk_no_automute	no	yes	no	0404	003c	006b
lin_codec-dump-spk_no_automute_dualstream	no	yes	yes	0404	001f	0079

Interpretation: Coeff 0x46 flips when the headset is plugged (dual), while 0x77 and 0x78 vary with automute and manual dualstream behavior. This aligns with the blog guidance that Windows/Linux differences often live in the vendor coefficient (verb) block rather than in the WID defaults.

1.5. Results

1.5.1. WID section stability

All 27 files contain the same 11 WID lines with identical values, indicating that the device-state verbs are **not** encoded by WID changes in this dataset.

WID	Codec	Drv	Loc
12	40000000	40000000	00000000
13	411111F0	411111F0	00000000
14	90170120	90170120	00000000
17	90170120	90170120	00000000
18	81D111F0	—	—
19	03A11030	03A11030	00080000
1A	411111F0	411111F0	00020400
1B	411111F0	411111F0	00000000
1D	40471A6D	411111F0	00000000
1E	411111F0	411111F0	00000700
21	03211010	03211010	00020000

1.5.2. Feature difference summary

Feature	Paired comparisons	Keys ≥ half pairs	Keys all pairs
preferred_device	3	15	8
headset_plugged	1	36	36
speaker_dolby	6	5	3
speaker_enhance	6	9	3
headset_dolby	3	10	10
headset_enhance	4	14	3

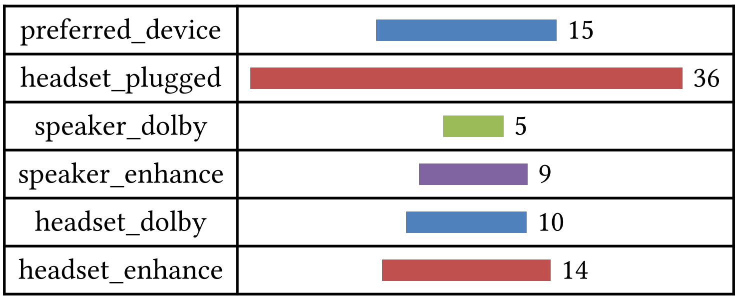

Table 1: Signature key counts by feature (≥ half of paired comparisons)

1.5.3. Preferred device signatures

Key	Coverage	Speaker preferred	Headset preferred

(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2	3/3	02 00 00 00 01 00 00 00 EF 06	02 00 00 00 01 00 00 00 EF 01
(REG_BINARY) Position	3/3	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4	3/3	41 00 00 00 01 00 00 00 01 00 01 00 01 0…0 00 40 34 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00
(REG_SZ) {24dbb0fc-9311-4b3d-9cf0-18ff155639d4},3	3/3	{0.0.0.00000000}.{35e064d4-3174-4fd5-ace4-5287bc0bdf22}	{0.0.0.00000000}.{105abd99-be3c-4986-856c-3f24ec337b55}
(REG_BINARY) {bb8bdb4a-edac-4660-9056-8e67e68e4e77},4	3/3	41 00 00 00 01 00 00 00 C7 1B 04 75 36 2…3 E0 37 6D 08 5A	41 00 00 00 01 00 00 00 0B B9 7A 91 18 0…9 1A 56 F8 04 5A

1.5.4. Headset plug signatures

Key	Coverage	Headset plugged	Headset unplugged
(REG_DWORD) InternalSpeakerStreamActive	1/1	0x0000	0x0001
(REG_BINARY) {5510c7ab-dfc2-40d0-a98b-5f77f697005e},2	1/1	03 00 00 00 01 00 00 00 28 00 00 00	03 00 00 00 01 00 00 00 00 00 00 00
(REG_BINARY) {5510c7ab-dfc2-40d0-a98b-5f77f697005e},1	1/1	03 00 00 00 01 00 00 00 08 00 00 00	03 00 00 00 01 00 00 00 02 00 00 00
(REG_DWORD) {3ba0cd54-830f-4551-a6eb-f3eab68e3700},6	1/1	0x0000	0x0001
(REG_BINARY) {194ef948-7cdb-403e-9f47-19418f7b248d},2	1/1	40 00 00 00 01 00 00 00 8D F3 96 28 D4 9A DC 01	40 00 00 00 01 00 00 00 59 36 9F D1 D3 9A DC 01

1.5.5. Speaker Dolby signatures

Key	Coverage	Speaker Dolby on	Speaker Dolby off
(REG_BINARY) Position	6/6	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4	6/6	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 60 23 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1	6/6	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 60 23 00 00
(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2	4/6	02 00 00 00 01 00 00 00 20 0C	02 00 00 00 01 00 00 00 EF 07
(REG_BINARY) {8a845654-d6c3-4cd7-b4eb-243d4bd99032},2	3/6	41 00 00 00 01 00 00 00 00 00 00 00 00 0…E F9 95 53 37 90	41 00 00 00 01 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00

1.5.6. Speaker enhancement signatures

Key	Coverage	Speaker enhancement on	Speaker enhancement off
(REG_BINARY) Position	6/6	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4	6/6	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1	6/6	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00
(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2	5/6	02 00 00 00 01 00 00 00 EF 01	02 00 00 00 01 00 00 00 EF 02
(REG_DWORD) {1da5d803-d492-4edd-8c23-e0c0ffee7f0e},5	4/6	∅	0x0001

1.5.7. Headset Dolby signatures

Key	Coverage	Headset Dolby on	Headset Dolby off
(REG_BINARY) {8a845654-d6c3-4cd7-b4eb-243d4bd99032},2	3/3	41 00 00 00 01 00 00 00 00 00 00 00 00 0…F E8 0F 4D 39 5D	41 00 00 00 01 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2	3/3	02 00 00 00 01 00 00 00 EF 07	02 00 00 00 01 00 00 00 EF 05
(REG_BINARY) Position	3/3	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4	3/3	41 00 00 00 01 00 00 00 01 00 01 00 01 0…0 00 80 34 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 C0 33 00 00
(REG_BINARY) {6737016f-5360-48ee-af05-e616c8ff27a7},2	3/3	02 00 00 00 01 00 00 00 04 00	02 00 00 00 01 00 00 00 00 00

1.5.8. Headset enhancement signatures

Key	Coverage	Headset enhancement on	Headset enhancement off
(REG_BINARY) Position	4/4	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4	4/4	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 24 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1	4/4	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 24 00 00

(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2	3/4	02 00 00 00 01 00 00 00 EF 01	02 00 00 00 01 00 00 00 EF 02
(REG_BINARY) {624f56de-fd24-473e-814a-de40aacaed16},3	2/4	41 00 00 00 01 00 00 00 FE FF 02 00 80 B…0 AA 00 38 9B 71	∅

1.6. Conclusion

The device-state verbs are reflected in **registry key/value differences**, not WID line changes. Preferred device and headset plug state show strong, consistent content signatures, while Dolby and enhancement states show fewer but still repeatable key changes across paired comparisons. The signature tables above provide the definitive, content-based mapping from each state to the specific keys that change.

1.6.1. Windows vs Linux implementation differences

- Windows: exposes separate speaker/headset endpoints and maintains independent streams. Registry (`REG_*`) deltas track preferred device, Dolby, and enhancement independently per endpoint.
- Linux: exposes a single logical playback device with shared stream IDs for speaker/headphone outputs (node 0x02/0x03). Automute and manual unmute are reflected in pin amp-out values (node 0x14/0x17) and Node 0x20 coefficients, not separate streams.

1.6.2. Consistency guidance

To make platform behavior consistent, choose one of these strategies:

1. Match Windows behavior on Linux by exposing separate sinks (speaker/headphone) via ALSA UCM or PipeWire/WirePlumber and routing distinct streams to each output, while disabling automute.
2. Match Linux behavior on Windows by forcing a shared stream (mirrored output) so speaker/headphone always play the same audio flow.

The analysis above indicates that Linux currently controls routing/mute state (pin amp-out values and Node 0x20 coefficients), while Windows controls endpoint selection and processing (registry keys). Aligning the control surface is the key to cross-platform consistency.

1.7. Appendix: Regeneration

Generate the CSV summary and the PDF report from the current dumps:

```
python rthddump_report.py --output rthddump_summary.csv
typst compile RtHDDump_Report.typ RtHDDump_Report.pdf
```